NeuralWave Hackathon:
Detecting Photovoltaic Installations from Smart-Meter Data

Dropped Neurons

1 Intro

1.1 Task Formulation

Given $\mathcal{C}$ set of customers, for each customer $i \in \mathcal{C}$ we observe *only* its metered electricity consumption time series $\boldsymbol{x}_i = (x_{i,t})_{t=1}^{T}$, sampled at a fixed resolution Δt. We think of the installed PV capacity of customer i as a non-decreasing step function

$$C_i(t) = \sum_{k=1}^{K_i} P_{i,k}\, \mathbf{1}[t \geq \tau_{i,k}]\,, \qquad \tau_{i,1} < \cdots < \tau_{i,K_i}, \tag{1}$$

where $K_i \in \mathbb{N}_0$ is the number of installations (the initial system plus later extensions), $\tau_{i,k}$ is the commissioning date and $P_{i,k} > 0$ the nominal capacity (kWp) of the k-th installation. The goal is a reproducible mapping $f : \boldsymbol{x}_i \mapsto \hat{C}_i(\cdot)$, which breaks down into:

1. **PV detection** (binary classification): $\hat{y}_i = \mathbf{1}[\hat{K}_i \geq 1] \in \{0, 1\}$;
2. **Commissioning date** (change-point detection): if $\hat{y}_i = 1$, estimate $\hat{\tau}_{i,1}$;
3. **Installed capacity** (regression): if $\hat{y}_i = 1$, estimate $\hat{P}_{i,1}$ in kWp;
4. **Extensions** estimate $\hat{K}_i$ and, for each $k = 2, \ldots, \hat{K}_i$, the pair $(\hat{\tau}_{i,k}, \hat{P}_{i,k})$.

1.2 Exploratory analysis

We started early on by understanding what information is retained in available data and establishing a problem baseline by implementing naïve models and adaptation from literature. Initial testing was done on fairly unreliable 30-10 split. Given how promising the unlabelled data looked, it was clustered in an hyperpace made of custom features based on day and night power consumption differences and ratios to predict task 1 classes. This approach has some obvious limits:

1. The performance is not on par with other methods applied nor the standard in literature
2. The feature hyperspace was taken from [3] that had a different dataset. However, optimizing it would require more time, data and, possibly, additional features.

Method	Accuracy	Paper Accuracy	Source
Vision Transformer (ViT)	Overfitting	Varying	Varying
CNN/MLP	0.89	0.90	[5]
Unsupervised clustering	0.87	0.99	[3]
Logistic regression	0.93	Varying	Varying
LSTM	0.61	Varying	Varying

Table 1: Report of literature methods compared to when adapted to our dataset

1.3 LLM and agents disclaimer, reproducibility

As data access was prohibited, agents received dataset sketches variables and with random numbers. AI-assisted coding was used to adapt paths and existing solutions. Seeds, package versions, and manual annotations are provided alongside the code.

2 Methodology

2.1 Time series as an image

After working on the data structure and testing multiple data representations, we found that reshaping each series into a matrix with one row per day and one column per time-of-day slot (e.g. 96 columns at 15-minute resolution), and rendering it as a heatmap (Figure 1), allowed for an extremely explicative and visually meaningful representation. In this view a PV system appears as a band of reduced (or negative) consumption around solar noon, with the band widening in summer and narrowing in winter, following daylight hours seasonality. It is very easy to spot how the commissioning date is the column where the sun-dependent signal band abruptly appears. This opened two possible paths, manual labeling of part of the unlabeled dataset and CNN/Vision/Transformer based models analisys. Consider that the signal is a negative effect on the consumed power and is therefore only visible when the customer power draw exceeds the generation and is otherwise zero-clipped.

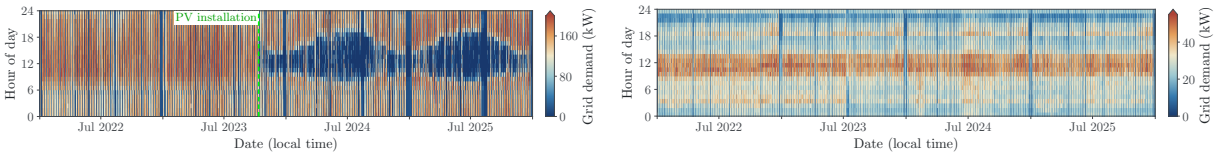


Figure 1: Consumption series reshaped as day × time-of-day images. PV generation appears as a midday band starting at the commissioning date and following the dailight hours seasonality.

2.2 Data and Constraints

With only 40 labelled development cases, supervised learning alone could easily overfit larger models so we made sure to use cross validation and few data specific approaches. One central idea of our approach is to turn the 1 148 unlabelled series into a large labelled dataset. It therefore became clear that to work under these constraints we had to leverage:

1. Simple models.
2. A way to get more data to train our model and still have reliable evaluation.

Instead of using external datasets our approach was to manually label part of the unlabeled data to train a high accuracy model necessary to avoid error propagation in the next step, which is to use that model to extend labeling to the whole dataset. To annotate a partition of the unlabelled dataset, three human annotators independently assigned each series a binary PV label, an estimated installation date, and a confidence score. A control sub-partition was annotated by all three annotators to measure agreement. We computed mean pairwise Cohen's κ [1], a standard inter-annotator agreement metric, obtaining $\kappa = 0.79$, indicating strong agreement. Disagreements were concentrated on a small number of genuinely ambiguous traces with noisy consumption profiles.

2.3 Task 1: PV detection

A lightweight CNN with approximately 84k parameters processes the manually annotated images using three convolutional blocks along the time-of-day axis, followed by a temporal layer that captures patterns across days. Dropout is used for regularization, and an attention head aggregates the daily representations into a single PV prediction for each customer. The convolutional backbone is first pretrained through masked-patch reconstruction and then fine-tuned

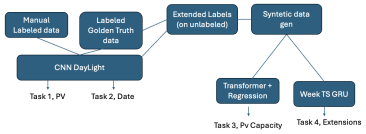


Figure 2: Pipeline overview

end-to-end using binary cross-entropy loss on both daily and customer-level labels.

2.4 Task 2: Date of installation prediction

For each day, a gradient-boosting classifier predicts a score based on a 16-dimensional self-supervised profile embedding and six engineered features. These include the midday-to-shoulder consumption ratio, the midday minimum, the daily mean, and sine and cosine encodings of the day of the year (from cyclical time encoding)[4]. The daily scores are smoothed using a seven-day moving average, and the installation date is estimated by detecting a single change point in the resulting sequence.

2.5 Task 3: Before/after pairs for capacity estimation

We combine two models: a Ridge regression that performs well on its own, and a Transformer-based neural network that improves predictions in more challenging cases. The Ridge model uses physical indicators of PV installation, while the neural network learns temporal patterns from daily consumption embeddings. The network combines a 1D-CNN for embeddings and a four-layer Transformer, which is pretrained on synthetic data derived from augmentation and chunking of time series before being fine-tuned on 22 labelled customers. The final kWp estimate is the geometric mean of the two models' predictions as utilizing both models empirically showed better results.

2.6 Task 4: Detecting extensions

As only one extension example was given, to detect system extensions we manually identified 11 high-confidence cases in the unlabelled pool, retaining only instances flagged by all three annotators. The PV extension model identifies when an existing solar installation has been enlarged, without confusing extensions with first-time installations. It processes sequences of weekly electricity CNN feature embeddings using a bidirectional GRU, considering load patterns, zero-withdrawal periods, data coverage and seasonality. It distinguishes between no change, a new installation and an extension, and flags likely events based on predicted probabilities. The event date is refined using changes in midday electricity withdrawal, while the increased PV capacity is estimated from year-over-year changes in the same seasonal period.

3 Results and Evaluation

It's worth noting that most of the failure modes for task 1 and especially task 2 were in fact very edge case scenarios or mismatching between the reported starting date with AIL and the actual start of the consumption.

Task	Metric	Result
PV detection	F1	0.98
Commissioning date	F1	0.61
Installed capacity	MAPE (kWp)	23%
Extension detection	F1	0.50

Table 2: Cross-validated results on the development set.

3.1 Future directions

A bottleneck is the scarcity of labelled data; additional annotations would give improved performance. We trained an attn based model on images (slightly lower accuracy vs cnn) which provides visually guided explainable data understanding which could be improved upon [see github]. Meteorological covariates would further help disentangle PV generation from household consumption, reducing noise in the signal. Finally, training on the timeseries with PODs locations, the PV arrays could be turned into a virtual distributed "camera" looking at the sky using Graph NN, giving a 15-min res. power model with cloud movement dynamics. [2].

References

[1] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20:37 – 46, 1960.

[2] Chandra Sekhar Charan Dande, Nikolaos A. Efkarpidis, Matthias Christen, Mirko Ginocchi, and Antonello Monti. Consumer phase identification in distribution grids using graph neural networks based on synthetic and measured power profiles. *Energy and AI*, 2025.

[3] Maomao Hu, Rory Telford, Dongjiao Ge, Bruce Stephen, and {David C.H.} Wallom. Classification and characterization of intra-day load curves of pv and non-pv households using interpretable feature extraction and feature-based clustering. *Sustainable Cities and Society*, 75, December 2021.

[4] Stefan Meisenbacher, Marian Turowski, Kaleb Phipps, Martin Rätz, Dirk Müller, Veit Hagenmeyer, and Ralf Mikut. Review of automated time series forecasting pipelines. *WIREs Data Mining and Knowledge Discovery*, 12(6):e1475, 2022.

[5] {Andres F.} {Moreno Jaramillo}, Javier Lopez-Lorente, {David M.} Laverty, Jesus Martinez del Rincon, {D John} Morrow, and {Aoife M.} Foley. Effective identification of distributed energy resources using smart meters net-demand data. *IET Smart Grid*, 5(2):120–135, February 2022.